Breast Cancer Diagnosis with Machine Learning : Evaluating Models on Wisconsin Dataset - fairness- Aware Perspective .

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Abstract :

Breast cancer is the most common disease among women worldwide, and it continues to pose a serious threat to global health [1]. Early detection will be crucial in maximizing the success of treatment and reducing death rates. Since the time artificial intelligence (AI) was in the limelight, machine learning (ML) algorithms have been found to enhance diagnostic accuracy. Earlier, their effectiveness cannot be questioned, yet these applications can inadvertently introduce bias, and discriminatory results for certain demographic groups though trained using very heterogeneous datasets [3][4].

This study meets the call for fairness in AI-based medical diagnosis by evaluating the prediction performance and fairness of three supervised machine learning (ML) models Logistic Regression(LR), Support Vector Machines(SVM), and Neural Networks (NN) applied to the Wisconsin Breast Cancer Diagnostic Dataset (WBCD) [2]. Given that the dataset lacks demographic attributes, we created synthetic racial group labels to simulate a binary race attribute. Fairness was assessed by rigorously studied fairness metrics: Demographic Parity and Equalized Odds [12][14] and was achieved with the Fairlearn library in python [18].

Our results show that while all models were extremely accurate (97–98%), slight gaps (2–4%) in fairness metrics were observed across racially simulated groups. Such findings indicate the need to include fairness testing in ML pipelines so that there can be fair diagnosis results in healthcare systems increasingly reliant on Ai .

1. introduction :

Breast cancer is one of the first diseases ever described, with records dating back as more than 3,500 years. Despite the progress of medical science, breast cancer remains one of the most prevalent and fatal forms of cancer worldwide. The World Health Organization (WHO) estimated that in 2022 around 2.3 million women were diagnosed with breast cancer, and this accounted for an estimated 670,000 fatalities globally [1].

In the last few years, Machine Learning (ML) has been a game-changer in the health care sector, offering new avenues for the detection of diseases, classification of them, and prognosis prediction. Among all its uses, early detection and classification of breast cancer have gained substantial attention since the potential to increase the accuracy of diagnosis and assist doctors in decision making at the correct moment exists. One of the most popular datasets to train and test such models is the Wisconsin Breast Cancer Dataset (WBCD) [2], which provides structured, labeled data that can be perfect for utilization with supervised learning methods.

Even as greater predictive accuracy has been the impetus for much of the work being conducted in this area, its ethical and social implications are becoming increasingly appreciated in the application of ML in healthcare. Particularly, concerns have been raised about algorithmic bias and its potential to produce unjust outcomes across population categories [3][4]. Biased algorithms could in effect sow or even increase existing health disparities in access to healthcare, diagnostic accuracy, and treatment effectiveness [5]. Such risks could manifest in the form of delayed diagnosis, increased misdiagnosis rates among underrepresented groups, and decreased trust in AI-driven medical systems—particularly among those that have historically experienced systematic health inequities in healthcare provided to them [6][7].

This research addresses the pressing issue of racial discrimination in ML-based breast cancer diagnosis. Our primary objective is to assess whether discrimination exists in predictive performance between racial groups and, more importantly, between simulated White and Black patient groups. Since we lack racial and demographic information in our baseline WBCD, we generated synthetic demographic labels through controlled random assignment so that we can reproduce racial subgroups without altering the statistical characteristics of the dataset. Using simulated subsets, we conduct a structured comparison of three widely used ML models : Support Vector Machine (SVM), Logistic Regression (LR), and Neural Networks (NN). Furthermore, we employ fairness evaluation metrics and examine bias mitigation methods to analyze and improve the equity of model outputs.

keywords : Machine learning (ML) ,Breast cancer , Demographic Parity,Equalized Odds, Support Vector Machine (SVM) , Logistic Regression (LR) and Neural Networks (NN) .

2. literature Review:

Machine learning (ML) has revolutionized medical diagnosis to a great extent by equipping it with tools capable of assisting detection, as well as prediction of diseases. The research mostly deals with present the performance of the models through selecting suitable algorithms and data preprocessing.

As these are attained, ML medical diagnosis fairness and bias remain to be investigated, particularly racial and demographic ones. ML fairness is stated to entail algorithm performance equalization to all the subgroups in the population. As trained with datasets that reflect biases deeply ingrained in society or in a system, the generated predictions bias against some populations—a not so desirable outcome in health care, where diagnosis decides life-altering outcomes.

There have been recent studies that have begun to clarify such issues. Obermeyer et al. (2019) [5], for instance, demonstrated how an average algorithm perennially underpredicted the medical need of Black patients as it used healthcare cost as a proxy for medical need. This resulted in colossal variability in treatment recommendations with Black patients being recommended 17.7% to 46.5% fewer treatments than White patients. Apart from that, Seyyed-Kalantari et al. (2021) [7] created that chest X-ray classification models demonstrated poorer diagnostic actions in Black patients than White patients. Another controversy-as-similar research is Buolamwini and Gebru (2018) [6], which uncovered staggering demographic differences in commercial facial recognition software with a 34.7% error rate in Black women but only 0.8% in White men.

In this research [19], Lui ,Y et al, they investigated how to make deep learning breast cancer detection models fair between racial and ethnic groups. Deep learning models such as ResNet-50 and Swin Transformer V2 were used to classify mammography images with good accuracy. A evident sign of discrimination in performance between racial and ethnic groups, and thus algorithmic bias was present. No removal of bias was used, but the research highlights the need to make machine learning systems fair".

Khan, S., et all" In this study [20] , resolved the issue of post-processing for debiasing in breast cancer staging using deep learning. The impact of post-processing methods like equal opportunity and equalized Odds on bias in breast cancer staging models was studied. Post-processing machine learning and deep learning methods like Naive Bayes, Neural Networks (NN) and Support Vector Machine "SVM" were explored. While fairness improved, post-processing alone was seen not to be adequate to eliminate bias.

In this paper [21] Joshi and Nayak provide a systematic review of Explainable Artificial Intelligence (XAI) approaches specifically toward the prediction and diagnosis of breast cancer risk. The authors critique a variety of XAI approaches, including SHAP (SHapley Additive exPlanations), Individual Conditional Expectation (ICE) plots, and surrogate models. These methods are emphasized for their ability to enhance the interpretability and trustworthiness of AI-driven diagnoses for breast cancer =. Although these XAI approaches show great promise, as per writers, factual evidence of their worth is in its beginning phase. This highlights the need for further research to establish the real-world application of XAI in the clinical setting and to ensure that these techniques will be able to consistently improve decision-making processes in breast cancer diagnosis.

In this study [22] Zhou,H and Liang,J (2024) explain the integration of convolutional neural networks such as VGG-16, Inception-V3 and ResNet with explainable artificial intelligence(XAI) techniques in the attempt to enhance breast cancer diagnosis. The authors note that the integration of such technologies not only improves diagnosis accuracy but also offers clinical practitioners an improved comprehension of AI-based decisions. With increased interpretability and cooperation between healthcare practitioners and AI systems, this method aims to enhance the process of diagnosis and establish trust in AI-assisted clinical practice.

In this paper [23] Iqbal et al. (2022), comparative evaluation of various machine learning algorithms for breast cancer classification was attempted, with consideration of algorithms such as BERT, LSTM, Naive Bayes, SVM, and Random Forest. Of the compared algorithms, BERT performed better with a 92.5% accuracy rate. The finding suggests BERT's potential as an effective diagnostic aid in text-based breast cancer diagnosis and entails its application in enhancing diagnostic accuracy in practice.

in their work [24] Germani et al.(2025) analyzed the bias and generalizability of breast mammography foundation models across a variety of datasets. The study aimed to explore performance variation across aspects such as breast patient age, and geographic region. Pre-trained models, domain adaptation, and fairness-aware training were used. Performance was evaluated using metrics such as AUC, and fairness calibration metrics for subpopulations. The research found multi-source data to improve performance but still contained pervasive biases in underrepresented populations. The paper points to the critical need for fairness-aware design during training and not post-hoc adjusting.

in this paper [25],Alom et al(2025). proposed an interpretable deep neural network model named DNBCD for accurate breast cancer diagnosis using histopathological and ultrasound images. The study was aimed at increasing the transparency and interpretability of AI models within clinical settings. With the integration of DenseNet121 and Grad-CAM for visual explanation, the authors advocated accuracy and interpretability. They trained the model with enriched and class-balanced data, wherein it achieved 93.97% and 89.87% accuracy on BreakHis-400x and BUSI datasets, respectively. The paper shows how it is possible to build clinical trust by combining high-performance models with explainability techniques, and suggests extending this approach to other imaging modalities in future applications.

in this research [26], Xu et al(2024) . conducted a comprehensive systematic review of fairness problems in deep learning-based medical image analysis .The objective was to categorize fairness evaluation approaches and mitigation techniques at pre-processing, in-processing, and post-processing. The review explained techniques such as GAN-based balancing of data, reweighting in training, and post-hoc correction methods such as equalized odds. Findings emphasized significant age and sex differences, and called for multi-phase models of fairness that are supported by representative and balanced data sets. It encouraged empirical studies that replicate real clinical settings to more strongly validate fairness interventions.

These findings underscore the need to test for the accuracy and fairness of ML models. Fairness metrics – such as Demographic Parity, which demands positive predictions to be equalized across groups, and Equalized Odds, which demands false positive and false negative rates to be equalized across groups are now standard to assess algorithmic fairness [3][4]. No such endeavor has ever been done in breast cancer detection, and it is the study that underlines a vast gap in the literature.

This study attempts to bridge this gap by investigating the fairness of certain ML breast cancer predictors from WBCD. Because race isn't an attribute of the data, artificial demographic cohorts will be formed so that racial imbalances can be created. Fairness and accuracy of such models will then be established with standard fairness measures. By doing so, the study aims to research the potential racial bias of AI-based breast cancer diagnosis and contribute its research to the overall discourse on unbiased healthcare AI.

3. Methodology:

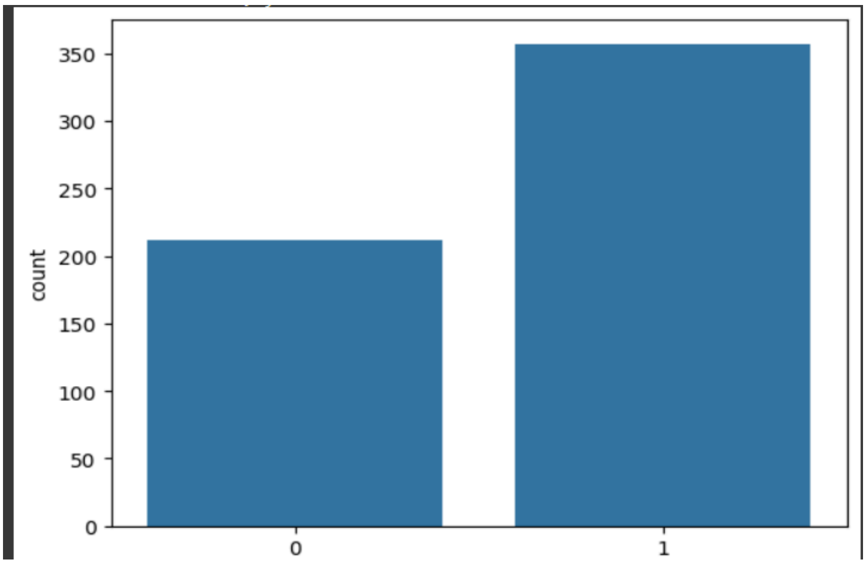
This section will explain the data set, what was done to fit the research objective, the models that were used, and what was taken to evaluate the performance of the models.

3.1 dataset :

in this research used the "Breast Cancer Wisconsin Dataset"(WBCD) from the sores UCI machine learning Repository[2]. We load a dataset from the scikit-learn library.This dataset is famous and often used for breast cancer research with machine learning.

It has information about different features of cell nuclei , this dataset includes 30 numerical features and 596 records and the dataset is generally divided into factors(features) called "x" and the outcome we will infer from them is called "y". Each sample in the dataset is labeled as either 'malignant' (cancerous) equal '0' or 'benign' (not cancerous) equal '1' in the dataset and the number of 'malignant' 212 and number of 'benign' 357.

The figure below distribution of each class in the two target classes:



` Figure (1).

3.2 Checking the Data:

First, in this research it looks at the data to see if any information was missing. it verified all the columns , all this in pandas library . We found that there was no missing information in our dataset.

3.3 Creating a Pretend Race Group (Sensitive Feature):

The original WBCD does not have information about the race of the patients. To study fairness between different race groups, we needed to add this information.

○ We created a new column called 'complexion'.

○ For each person (row) in the dataset, we randomly assigned them to be either 'white' or 'black'. We made sure this was done randomly so that the groups were about the same size. This 'complexion' column is our "sensitive feature" because it is the feature we will use to check for fairness , So we have this number of features (31).

The figure below shows the feature (2) sensitive feature ,It turns out that the two races are close .

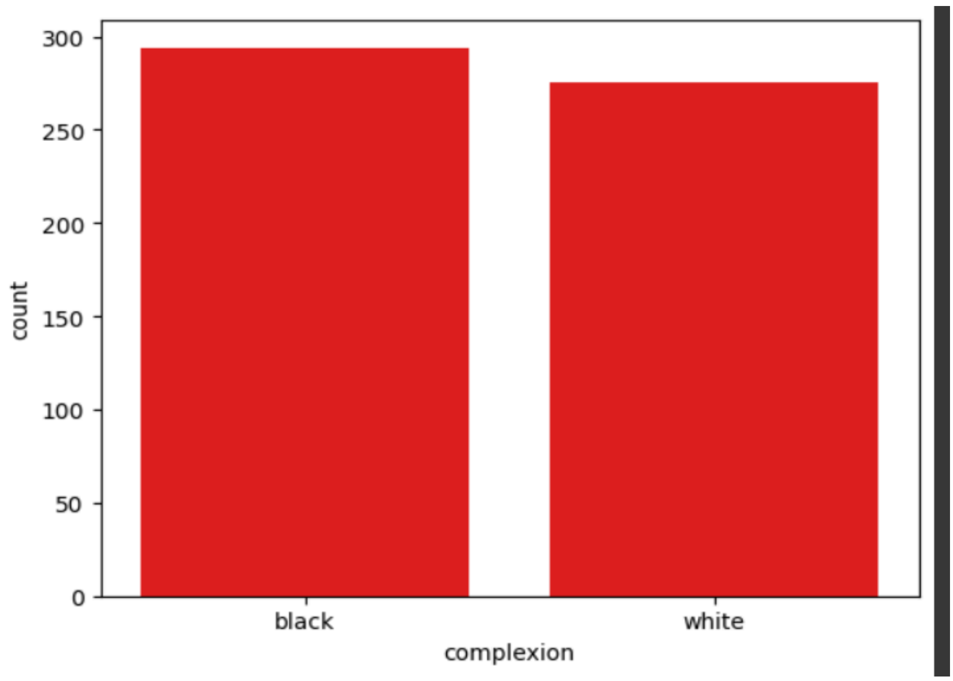


figure (2).

3.4 Changing Words to Numbers (Encoding):

Machine learning models usually work better with numbers than with words ,
like Neural Networks (NN) and Support Vector Machine (SVM) .

- ○ So, we changed the 'complexion' column. We made 'black' equal to the number 1 and 'white' equal to the number 2.
- ○ We then removed the old 'complexion' column with words. We now have a column called 'complexion_encoded' with numbers for Avoidance is a recurring theme..

3.5 Preparing the Data for Models (Scaling):

The different features in the dataset (like 'mean radius' or 'mean texture') have numbers in different ranges. For example, one feature might have small numbers, and another might have big numbers. This can sometimes make it hard for the ML models to learn well.

it uses one of the most popular types of scaling, which is Standardization using Z-score. This makes the values have a mean of 0 and a standard deviation of 1, which is an ideal situation [8].

Mathematical Formulation :

$$X_i = \frac{X_i - \mu}{\sigma}$$

So, we used a technique called '**StandardScaler**'. This changes all the feature values so they are in a similar range. We did this for all features except our 'complexion_encoded' column .

3.6 Splitting the Data:

We needed to divide our data into two parts:

○ **Training data** : This is the data we use to teach the ML models (75% of the whole data).

○ **Testing data:** This is the data we use to see how well the models learned. We don't show this data to the models when they are learning (25% of the whole data). When we split the data, we also made sure to split our 'complexion_encoded' feature into training and testing parts.

3.7 the used Models :

it was used 3 machine learning algorithm :

1. **Logistic Regression [9]:** A linear model for binary classification that predicts the probability of the class using a logistic function

   Mathematical Formulation :

$$p(y = 1|X) = \frac{1}{1 + e^{-(\beta 0 + \beta TX)}}$$

where :

- **P(y=1∣X): Probability of class.**
- **β0: Intercept term,**
- **β^T X: Dot product of coefficients β and features X .**

2 . Support Vector Machine [10] : Supervised learning algorithm used for classification, It finds the optimal hyperplane that maximally separates the class in the feature space.

Key Concepts:

• **Hyperplane:** A decision boundary that separates the classes.

• **Margin:** The distance between the hyperplane and the nearest data points (support vectors).

• **Kernel Trick:** Allows SVM to handle non-linear decision boundaries by mapping data into a higher-dimensional space.

Mathematical Formulation :

$$\boldsymbol{Minimize}: \left(\frac{1}{2}\right) \parallel \boldsymbol{w} \parallel^2 + \boldsymbol{C}\ \boldsymbol{i = 1\sum n\xi i}$$
$$\boldsymbol{Subject\ to}: \boldsymbol{yi(wT\phi(xi) + b) \geq 1 - \xi i, \xi i \geq 0}$$

where :

• **ww:** Weight vector of the hyperplane.
• **CC:** Regularization parameter controlling the trade-off between margin size and classification error.

• ξiξi: Slack variable for handling misclassifications.

• **ϕ(xi)ϕ(xi):** Kernel function mapping input xixi to a higher-dimensional space.

3 . Neural Network (NN) (specifically MLPClassifier): A model inspired by how the human brain works, with layers of "neurons."

Mathematical Formulation :

$$z_j^{(l)} = \left(\sum_{i=1}^{n_{l-1}} W_{ji}^{(l)} a_i^{(l-1)} \right) + b_j^{(l)}$$

[11] **:** $W_{ji}^{(l)}$ **:** It is the name that relates the relationship between i in Canadian $l - 1$and neuron j in Canadian l.

$a_i^{(l-1)}$ **:** It is the output of neuron i of the previous layer $l - 1$.

$b_j^{(l)}$ **:** It is biased towards neuron j in layer j .

then will be update for whagits :

Mathematical Formulation :

$$W_i = W_i + a * [T - A] * X_i$$

- Where W_i is the vector of weights, X_i is the input vector presented to the network, T is the correct result that the neuron should have shown, A is the actual output of the neuron, alpha [0,1] is the training rate, and b is the bias.

Each model from these was evaluated on (accuracy ,precision, recall, F1- score). and we used demographic parity diff and equalized odds diff from the fairlearn library that will allow us to assess both overall performance and fairness across the synthetic racial groups.

3.8 . Fairness (Fairness Metrics): demographic parity diff and equalized odds diff.

1. **demographic parity [12][13] :** The goal is to ensure that the algorithm's prediction results are equal and unbiased across different groups (such as skin color).

Mathematical Formulation :

$$P(\hat{Y} = 1 | A = a) = P(\hat{Y} = 1 | A = b) \ \forall a, b$$

where :

$\hat{Y}$ **:** predict result from model (1 or 0).

A **:** sensitive feature (complexion)

2. **Equalized Odds [14]** : The goal is to ensure that the predicted results are unbiased with respect to the actual results, which is Y (target).

Mathematical Formulation :

$$P(\hat{Y} = 1 | Y = y, A = a) = P(\hat{Y} = 1 | Y = y, A = b) \ \forall \ y, a, b$$

Where :

Y **:** is the actual result .

- **This table "Table (1)" explains the difference between demographic parity and equalized odds.**

Table (1)

the standard	**Demographic parity**	**Equalized Odds**
focus	the results are equal "regardless of the truth"	performance equals "in relation to truth"
Sensitivity to performance	do not take into account the accuracy of the model for the groups	considers the accuracy of the model for the groups
Optimal use	when the political facts are not available	when the basic truth is known

4. Experiment :

After selecting the dataset and perprocess it (Checking and scaling) then choose Machine Learning (ML) algorithm , the next step is training the model then evaluating and applying Fairness (Fairness Metrics) .

4.1 : training and testing Model :

As we said section **3.6** we As we said, we separated the data into 75% training and 25% testing. The Machine Learning algorithms were selected for training (mentioned in section **3.7**).For The Logistic Regression(LR) and Neural Network (NN) (specifically MLPClassifier) have (1000) iterations .

The figure below shows mean for **(accuracy ,precision, recall, F1, demographic parity diff and equalized odds diff) :**

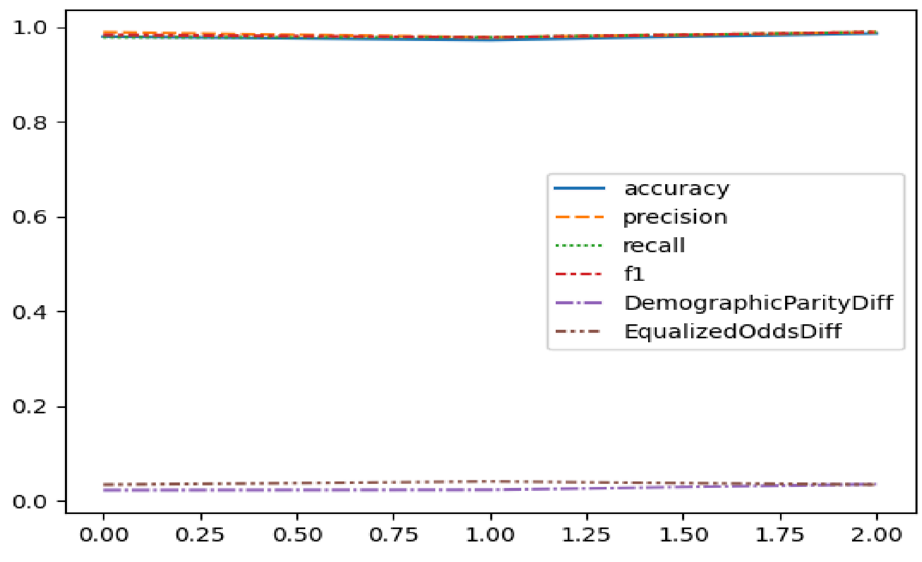


figure (3).

○ For each model, we used the training data (x_train, y_train) to teach it.

○ After training, we used the testing data (x_test) to make predictions about whether the cells were cancerous or not (y_pred).

4.2 : Evaluation section :

After training the model , we should evaluate the performance of the model by confusion metrics and Fairness Metrics .

- **accuracy [15] :** This evaluation metric is measuring the proportion of correct predictions (True Positives and True Negatives) among the total predictions made.

formula $$Accuracy = \frac{TP + TN}{TP+TN+FP+FN}$$

where :

1. TP**:** Correctly predicted positive instances.
2. TN**:** Correctly predicted negative instances.
3. FP**:** Incorrectly predicted positive instances.
4. FN **:** Incorrectly predicted negative instances.

- **Precision [16]:** This evaluation metric measures the accuracy of positive predictions, indicating how many of the predicted positives are positive.

Formula $$Precision = \frac{TP}{TP + FP}$$

Where :

- TP**:** Correctly predicted positive instances.
- FP**:** Incorrectly predicted positive instances.

- **Recall [16]:** This evaluation metric measures the ability of a model to find all positive instances.

Formula $$Recall = \frac{TP}{TP + FN}$$

Where :

- TP**:** Correctly predicted positive instances.
- FN **:** Incorrectly predicted negative instances..

- **F1-Score [16]:** This evaluation metric is the harmonic mean of precision and recall, providing a balanced measure of both.

Formula $$F1 - score = 2 * (\frac{Precision * Recall}{Precision + Recall})$$

Where :

- $Recall \; = \; \frac{TP}{TP + TN}$
- $Precision \; = \; \frac{TP}{TP + FP}$
- TP**:** Correctly predicted positive instances.
- FP**:** Incorrectly predicted positive instances.
- FN **:** Incorrectly predicted negative instances.

- **Fairness (Fairness Metrics):** demographic parity diff and equalized odds diff .

  Selection Rate: For each group ('white' and 'black'), how often the model predicted "cancer".

  Demographic Parity Difference: This measures the difference in the selection rate between the two groups. A small difference means the model is more fair by this measure.
  Equalized Odds Difference: This checks if the model makes mistakes (like missing a cancer or wrongly saying there is cancer) at similar rates for both groups. A small difference here also means more fairness.

  Note : this was mentioned more in **section 3.8** .

5. Results

In this section, there will be a description of the results of the algorithms that were used : Support Vector Machine , Logistic Regression , Neural Networks , after the training .Each model was evaluated on the same dataset and it has taken by (accuracy, precision, recall, F1-score, and the fairness metrics) to evaluated.In addition, matrices heatmap were presented to explain the performance of each model in terms of (accuracy, precision, recall, F1-score, and the fairness metrics).The analysis highlights both the predictive power and fairness of each model , offering a comprehensive view of their strengths and limitations in the context of equitable breast cancer diagnosis.

1. **Logistic Regression (LR) :**

   - Results :

     • **Accuracy**: The Logistic Regression classifier achieved an accuracy of 97% indicating its ability to classify instances correctly.

     • **Precision:** The precision score of 98% reflects the model's ability to minimize false positives.

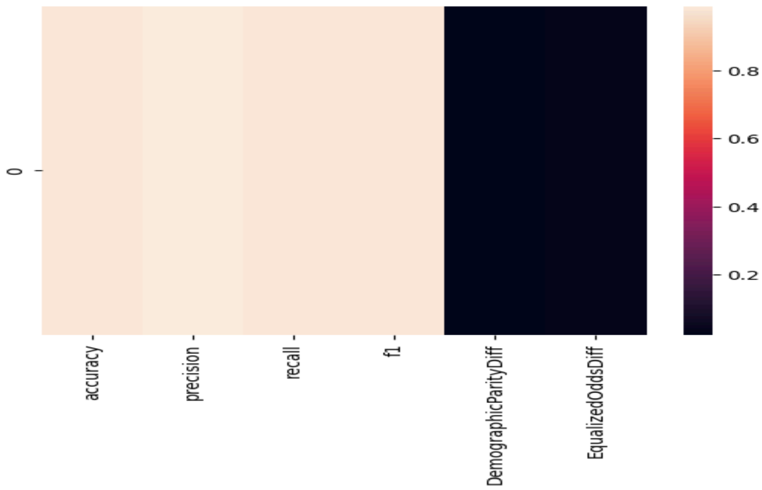
     • **Recall:** The recall score of 97% demonstrates the model's effectiveness in identifying true positives.

• **F1-Score:** The F1-score of 98% represents a balanced performance between precision and recall.

• **Demographic parity:** The demographic parity score of 2% reflects a slight bias in the final results, which is an acceptable indicator.

• **Equalized Odds:** The equalized Odds score of 3% reflects the difference in the model performance between groups (in terms of actual outcomes) .

- **The Metrics Heatmap :**



figure(4).

2. **Support Vector Machine (SVM) :**

    - Results :

• **Accuracy**: The Support Vector Machine classifier achieved an accuracy of 97% indicating its ability to classify instances correctly.

• **Precision:** The precision score of 97% reflects the model's ability to minimize false positives.

• **Recall:** The recall score of 97% demonstrates the model's effectiveness in identifying true positives.
• **F1-Score:** The F1-score of 97% represents a balanced performance between precision and recall.

• **Demographic parity:** The demographic parity score of 2% reflects a slight bias in the final results, which is an acceptable indicator.

• **Equalized Odds:** The equalized Odds score of 4% reflects the difference in the model performance between groups (in terms of actual outcomes) .

- **The Metrics Heatmap :**

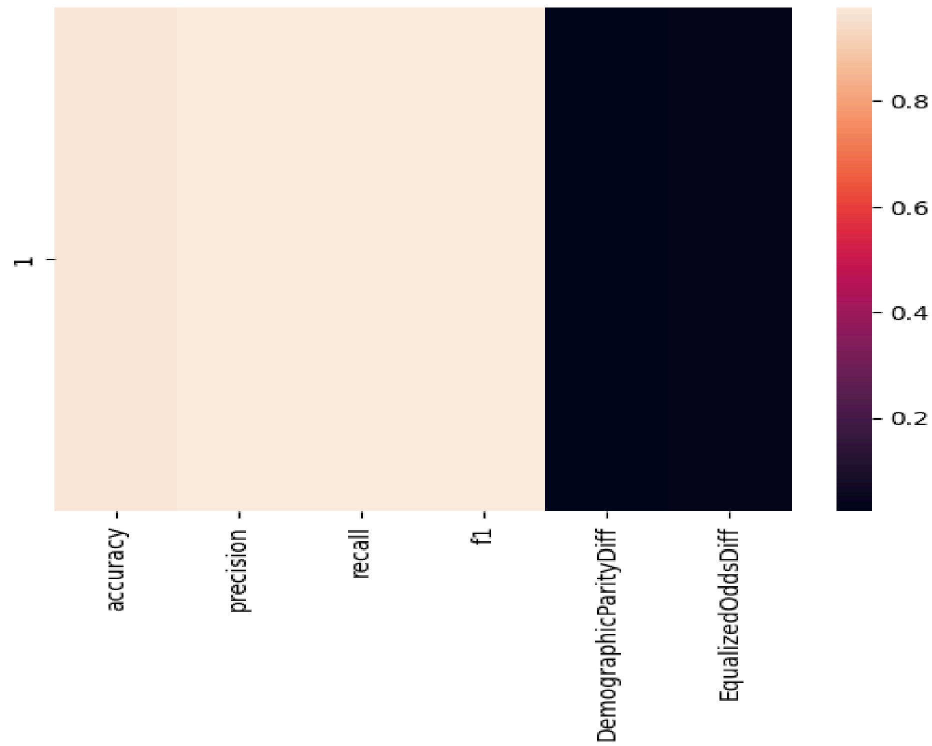


figure (5).

3. **Neural Network (NN) (specifically MLPClassifier) :**
   - Results **:**

     • **Accuracy**: The Neural Network classifier achieved an accuracy of 98% indicating its ability to classify instances correctly.

     • **Precision:** The precision score of 98% reflects the model's ability to minimize false positives.

     • **Recall:** The recall score of 98% demonstrates the model's effectiveness in identifying true positives.

     • **F1-Score:** The F1-score of 98% represents a balanced performance between precision and recall.

     • **Demographic parity:** The demographic parity score of 3% reflects a slight bias in the final results, which is an acceptable indicator.
     • **Equalized Odds:** The equalized Odds score of 3% reflects the difference in the model performance between groups (in terms of actual outcomes) .

- **The Metrics Heatmap :**

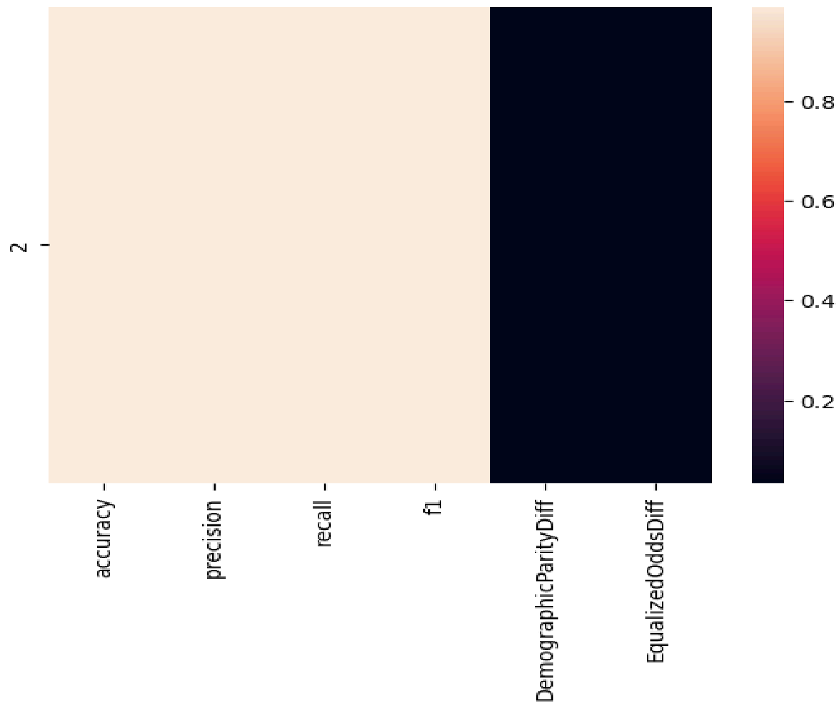


figure (6).

The figure below shows the result for performance each model in barchart figure :

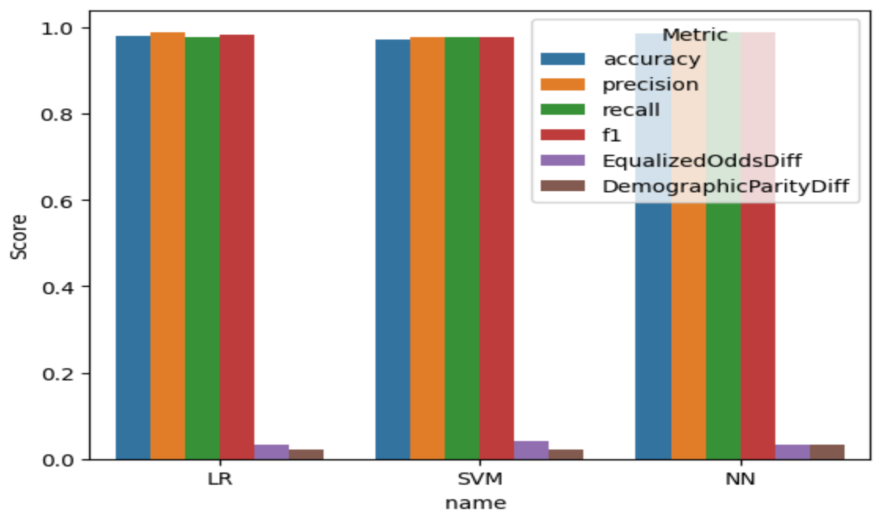


figure (7)

Finally, we put all the results (accuracy, precision, recall, F1-score, and the fairness metrics) into a table, called "Table (2)". This helps us compare how each model performed overall and how fair it was to the different groups.

Table (2)

Name	Accuracy	Precision	Recall	F1-Score	Demographic Parity	Equalized Odds
SVM	0.972028	0.977528	0.977528	0.977528	0.022421	0.040000
LR	0.979021	0.988636	0.977528	0.983051	0.021825	0.033333
NN	0.986014	0.988764	0.988764	0.988764	0.034325	0.033333

The figure below shows the linear regression of(accuracy, recall,precision,F1) for models:

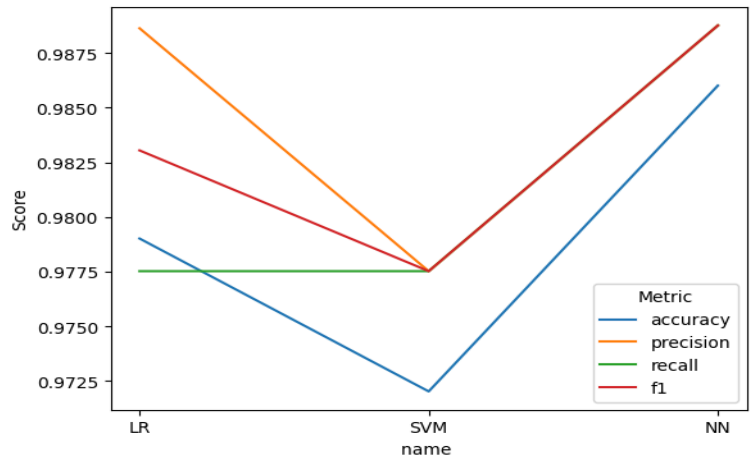


figure (8).

The figure below shows the linear of fairness metrics(Demographic Parity ,Equalized Odd):

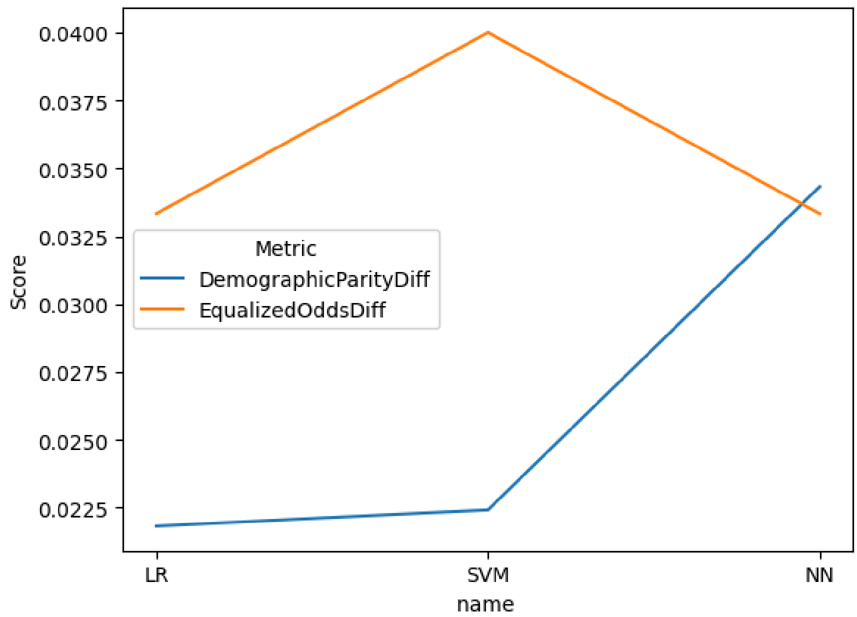


figure (9).

6. Discussion :

This section addresses the challenges and limitations we faced during the preparation of our research. These limitations are divided into the small sample size of the data,a lack of racial attributes,a trade-off between performance and equity and a lack of true demographic diversity.

6.1 dataset challenges :

One of the biggest existing problems is the lack of an ethnic attribute, as the 'WBCD' dataset does not contain ethnic information. We have simulated it with a 'complexion' attribute randomly selected from two categories (Black, White) to make it appear as if we collected individuals randomly (this method simulates potential bias scenarios). Additionally, the size of the dataset was relatively small, with a total of 596 cases.

6.2 Fairness VS Accuracy :

Although neural networks (NN) and Support Vector Machine (SVM) achieved better performance metrics, Logistic Regression (LR) resulted in a reduction in the variance of fairness metrics, indicating a balance between accuracy and fairness. Therefore, this emphasizes the importance of not relying solely on accuracy when using ML models in the medical sector, as it is a sensitive field and this can negatively impact the population.

7. Limitations and Moral Risk:

The biggest limitation is the lack of racial data, so we performed this artificially. indicates that realistic data must be provided in the future. Since artificial intelligence is widespread in the world, especially in the medical field, and people have become very dependent on it, scientists must focus on the direction of justice and reveal bias to take into account the human aspect and avoid problems of racism. Also, due to the limited time, it was possible to study the solution to this problem more deeply and to address more algorithms in machine learning, but it was sufficient to focus on detecting bias to increase awareness of the issue of justice.

8. Recommendation :

- **Comprehensive Diversification**: A conscious effort must be made to collect a comprehensive data set that reflects the diversity of human populations in terms of demographics. This requires the efforts of multiple healthcare institutions and in diverse regions.

- **Balanced representation:** Instead of relying solely on available data , efforts should be directed toward ensuring minority group representation in society that often represents the minority percentage in the current data. This can be achieved by collecting data or using data augmentation techniques carefully to create artificial examples that represent this group.

- **Collaboration with field experts:** Data collection and classification should be carried out in close consultation with physicians and clinical experts to ensure the accuracy and clinical relevance of the data and to avoid, as much as possible, introducing biases and unintended errors that may occur.

- Apply advanced bias mitigation techniques pre-processing (fairness-aware generalization), during processing (loss function or fairness constraints), and post-processing (equalized odds or probabilistic calibration), and continuously monitor and evaluate bias periodically.

9. Conclusion :

This research focused on analyzing fairness and detecting racial bias ratios between (white, black) in breast cancer diagnosis using Fairness Metrics: demographic parity diff and equalized odds diff. On machine learning algorithms (supervised learning) Neural Network (NN), Support Vector Machine (SVM),Logistic Regression (LR) on the WBCD dataset. It was found that The model with the highest accuracy and best demographic parity was the Neural Network (NN), while the model with the highest Equalized Odd was Support Vector Machine (SVM.) .

It was found that all models had a bias percentage in general ranging approximately between (0.02 - 0.04).Therefore, multidisciplinary collaboration must be encouraged to achieve a deeper understanding of clinical justice, and all levels of society must be brought together so that equality is achieved across society and racial problems are avoided this scientific research helps focus on reducing bias rates..

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